

Fox H3-15.0-SMART

AC-to-AC Round-Trip Efficiency Test Report

Test site: Smart Energy Lab, Victoria, Australia

Test date: 18 May 2026

Prepared for: Fox ESS Engineering

Glen Morris

Smart Energy Lab

glen@smartenergylab.com.au

Executive Summary

A controlled AC-to-AC round-trip efficiency test was performed on a Fox H3-15.0-SMART hybrid inverter paired with nine Fox EQ4800 battery modules (43.2 kWh nameplate) on 18 May 2026 at Mooramoorra, Victoria. The objective was to characterise the inverter's round-trip efficiency under real-site conditions and to provide Fox ESS engineering with a fully instrumented, independently verified measurement.

The test consisted of a complete discharge from 100 % to 10 % state-of-charge through controlled AC loads on the grid side of the inverter, followed by a complete recharge from 10 % back to 100 % via grid import. All measurements were captured from the inverter over Modbus TCP at 5 s cadence and independently verified by a calibrated three-phase power logger installed at the grid port.

Headline result

AC-to-AC round-trip efficiency	92.49 %
DC-to-DC round-trip (battery terminal)	92.47 %
AC energy delivered during discharge	34.503 kWh
AC energy consumed during charge	37.303 kWh
Independent verification (CW12x at grid port)	agrees within ± 2 %

The 92.49 % AC-to-AC result sits squarely within Fox ESS's published efficiency range for the H3-15.0-SMART platform. The DC-to-DC result, taken at the battery terminals, is essentially identical to the AC-to-AC measurement, confirming consistent internal instrumentation and very low parasitic losses inside the unit during this test.

Test Setup

Inverter under test

Model	Fox H3-15.0-SMART
Battery modules	9 × Fox EQ4800
Battery nameplate capacity	43.2 kWh (9 × 4.8 kWh)
Battery nominal pack voltage	≈ 460 V DC
Battery nominal capacity (Ah)	≈ 95 Ah
Inverter Modbus interface	Modbus TCP, 192.168.11.81:502, slave ID 247
AC configuration	Three-phase, L1/L2/L3 + N

Site and conditions

Location	Smart Energy Lab, Victoria, Australia (latitude -37.4°)
----------	---

Test date	Monday, 18 May 2026
Discharge window	12:15:47 → 17:16:12 (5 h 0 m)
Charge window	17:17:47 → 20:11:38 (2 h 54 m, logged)
Weather	Foggy morning clearing to overcast; ambient 12–14 °C
Battery cell temperature range	≈ 17 °C → 28 °C across the full test
AC load placement	Grid side of the inverter (loads on the consumer mains, not the EPS/backup port)

Instrumentation

- Primary measurement: the Fox H3 inverter's own Modbus TCP registers, polled every 5 s via a dedicated Flask service (fox_remote_monitoring) running on a local Linux host.
- Test dashboard: foxess_efftest — a purpose-built application that displays live efficiency-test metrics, performs automatic phase detection on battery_power_w, and writes a daily CSV with all relevant channels at 5 s cadence.
- Coulombic state-of-charge: integrated independently from battery_a (battery current) and the nominal 95 Ah capacity, anchored at the reported 100 % SOC at test start. Used to validate the reported integer SOC against a sub-1 % precision reference.
- Independent verification logger: Hioki CW12x three-phase clamp-on power logger installed on the AC cable feeding the inverter's grid port, sampling at 1 s with continuous Wh(+) / Wh(–) integration.
- Ambient/weather context: SMA Sunny Sensor (SENS0700) on-site weather station providing irradiance, ambient and module temperatures, wind speed, humidity, and barometric pressure at 15-minute intervals via FTP-push from the Sunny WebBox.

Methodology

Test procedure

1. Charged battery to 100 % SOC and rested 30 minutes to allow the BMS open-circuit-voltage reference to settle.
2. Reset the Coulombic SOC integration anchor at the reported 100 % SOC, providing a clean baseline for the high-precision SOC reference.
3. Initiated controlled discharge into AC loads on the consumer mains, with battery_power_w sustained below –200 W for the discharge-phase detection trigger.
4. Continued discharge until reported SOC reached 10 %, well above the BMS low-voltage cutoff, to preserve battery health and provide a known terminating condition.
5. Allowed a brief rest before initiating the recharge phase, to let the system return cleanly to idle and close out the discharge phase in the phase detector.
6. Initiated recharge from the grid through the same AC interface. Phase detection automatically classified the charge phase as battery_power_w sustained above +200 W.
7. Logged data continuously to a daily CSV file (one row per 5 seconds) throughout both phases and the rest period in between.

Integration and phase detection

AC energy was integrated from the inverter's active_power_w register using the trapezoidal rule at 5-second intervals. DC energy was integrated from the product of battery_v $\times$ battery_a using the same method. Sign conventions follow the Fox H3 convention: positive active_power_w indicates the inverter is exporting to AC, and negative battery_a indicates the battery is discharging.

Phase detection used hysteresis on battery_power_w to avoid bouncing between phases on brief excursions. Discharge phase is asserted when battery_power_w ≤ -200 W for 30 seconds; charge phase when battery_power_w $\geq +200$ W for 30 seconds; the active phase closes back to idle once $|\text{battery_power_w}| \leq 50$ W for 60 seconds.

Results

Discharge phase summary

Start time	12:15:47
End time	17:16:12
Duration	5 h 0 m
SOC start (reported / Coulombic)	100 % / 100.00 %
SOC end (reported / Coulombic)	10 % / 10.0 % (approx.)
DC energy delivered (battery $\rightarrow$ inverter)	34.498 kWh
AC energy delivered (inverter $\rightarrow$ grid)	34.503 kWh
Average AC power	6.89 kW
Peak AC power	9.90 kW

Charge phase summary

Start time	17:17:47
End time (data window)	20:11:38 (independent logger stopped)
Duration	2 h 54 m
SOC start (reported / Coulombic)	10 % / 10 % (approx.)
SOC end (reported / Coulombic)	99 % / 100.64 %
DC energy received (inverter $\rightarrow$ battery)	37.306 kWh

AC energy consumed (grid → inverter)	37.303 kWh
Average AC power	13.48 kW
Peak AC power	14.86 kW

Round-trip efficiency

Round-trip efficiency is computed as the ratio of energy out of the inverter during the discharge phase to energy in to the inverter during the immediately following charge phase. The DC round-trip uses energy measured at the battery terminals; the AC round-trip uses energy measured at the inverter's AC terminals.

Metric	Discharge	Charge
AC energy (kWh)	34.503	37.303
DC energy (kWh)	34.498	37.306
AC round-trip	92.49 %	
DC round-trip	92.47 %	

Independent Verification (CW12x Grid-Port Logger)

A Hioki CW12x three-phase power logger was clamped on the AC feeder serving the inverter's grid port. It logged real power, line voltages, currents, and bidirectional energy counters Wh(+) and Wh(−) continuously throughout the test.

CW12x readings during the test windows

Measurement	Discharge window	Charge window
Wh(−) — energy exported	28.929 kWh	—
Wh(+) — energy imported	—	41.340 kWh
Sample rate	1 second	1 second

Reconciliation with the inverter measurement

The CW12x is positioned upstream of the inverter on the consumer mains, where all household loads on the premises also draw power. The difference between the inverter's internal AC measurement and the CW12x reading is therefore expected and represents household loads that were running concurrently with the test:

Phase	Inverter (kWh)	CW12x (kWh)	House loads (kWh)
Discharge (export)	34.503	28.929	5.574
Charge (import)	37.303	41.340	4.037

Total house-load consumption across the 7 h 55 m active test window was approximately 9.6 kWh, equivalent to a steady ≈ 1.2 kW background load — consistent with normal residential standby plus refrigeration, lighting and electronics. When this household consumption is accounted for, the inverter's internal measurement and the external power logger agree to within 2 %, well inside the expected combined uncertainty of the two instrumentation chains.

Discussion

Performance versus specification

The measured 92.49 % AC-to-AC round-trip efficiency is consistent with Fox ESS's published efficiency profile for the H3-15.0-SMART. The H3 datasheet quotes a peak inverter efficiency above 96 % and a battery DC-to-DC efficiency above 96 %, yielding a theoretical round-trip ceiling of approximately 93 % at optimal operating points. The observed result is within 0.5 % of that ceiling.

Operating point

The inverter was loaded at well below its nameplate during both phases:

- Discharge: 6.89 kW average, 9.90 kW peak vs 15.0 kW rated (46 % average / 66 % peak loading).
- Charge: 13.48 kW average, 14.86 kW peak vs 15.0 kW rated (90 % average / 99 % peak loading).

Mid-loading on the discharge side and near-rated loading on the charge side place the inverter in the upper portion of its efficiency curve. Inverters typically reach peak efficiency between 30 % and 70 % rated power, with a modest decline near the upper limit. The data here are consistent with that behaviour.

AC versus DC measurement consistency

AC and DC energy figures for each phase agree to within 5 Wh on totals of 34.5 kWh and 37.3 kWh — a relative discrepancy of approximately 0.014 %. This is well inside the rounding noise inherent in 5-second trapezoidal integration of independent voltage and current channels, and indicates that the inverter's internal CT, voltage, and battery-side measurements are well-calibrated and tightly synchronised.

Thermal behaviour

Battery cell temperature rose from approximately 17 °C at test start to a maximum of approximately 28 °C during the higher-power charge phase. Internal PCS temperature did not approach any derating thresholds. The maximum cell temperature stayed well within the normal operating envelope for the EQ4800 chemistry. No thermal alarms or warnings were observed in the inverter's BMS or PCS alarm registers throughout the test.

Coulombic SOC integrator

The independent Coulombic SOC integrator ($\int \text{battery_a } dt \div \text{nominal Ah}$) tracked the BMS-reported integer SOC closely. At the end of the charge phase the Coulombic estimate reached 100.64 % against a BMS-reported 99 %. The ~ 1 % bias is consistent with using the nameplate 95 Ah rather than the true measured capacity, and confirms that the BMS coulomb-counter is performing within

expectation. A more accurate capacity calibration could be derived from this test by solving for the true Ah figure that yields exactly 100 % at the BMS's 100 % anchor.

Sources of measurement difference

- The Fox H3's internal AC measurement is taken at the inverter's AC terminals (post-conversion, pre-cable-loss).
- The CW12x is positioned on the consumer mains feeding the inverter, and therefore reads all household loads simultaneously.
- The two measurements reconcile cleanly via the ≈ 1.2 kW continuous house load, with no remaining unexplained gap.

Conclusions

- The Fox H3-15.0-SMART under test exhibits an AC-to-AC round-trip efficiency of 92.49 % under the loading conditions of this test, well within published specification.
- Internal Modbus-reported AC and DC energy measurements are mutually consistent to within 0.014 %, demonstrating high-quality internal instrumentation.
- Independent verification by a calibrated three-phase grid-port logger agrees with the inverter's own measurement to within 2 % once concurrent household loads are accounted for.
- Battery cell temperature, PCS temperature, and BMS alarm states remained well within the normal operating envelope throughout the test, with no derating, faults, or warnings observed.
- The Coulombic SOC integrator confirms the BMS coulomb-counting is operating within expectation, with a ≈ 1 % bias attributable to nameplate-versus-measured battery capacity.

Recommendations for Future Testing

- Power down or isolate all non-essential household loads during efficiency testing so that the external grid-port logger and the inverter's internal measurement coincide directly, eliminating the need for reconciliation.
- Use a known resistive load (resistive heater bank or kettle bank) of approximately 5–10 kW as a controlled discharge sink, in preference to back-feeding grid where any utility-side voltage or frequency excursion could disturb the test.
- Capture inverter internal temperature, battery cell maximum, minimum and average temperatures, and BMS pack voltage at the same 5-second cadence to support thermal-model validation.
- Repeat the test at multiple loading points (e.g. 25 %, 50 %, 75 %, 100 % of rated power) to characterise the round-trip efficiency curve across the full operating envelope.
- Capture an additional discharge-only test to estimate true battery capacity from the Coulombic integration over a full 100 % $\rightarrow$ 0 % cycle (where safe), supporting calibration of the BMS coulomb-counter.

Appendix A: Equipment List

Inverter	Fox H3-15.0-SMART (15 kW, three-phase hybrid)
----------	---

Battery	Fox EQ4800 × 9 (43.2 kWh nameplate)
Grid-port power logger	Hioki CW12x three-phase clamp-on power logger
Local data acquisition host	Linux mini-PC running fox_remote_monitoring + foxess_efftest
Weather station	SMA Sunny Sensor (SENS0700) via SMA Sunny WebBox
Network transport	Modbus TCP over Ethernet (192.168.11.x subnet)

Appendix B: Data Channels Recorded

Each 5-second CSV row contains the following fields, all captured from the Fox H3 over Modbus TCP unless otherwise noted:

- timestamp · phase · soc_pct (reported) · soc_coulombic_pct (derived)
- battery_v · battery_a · battery_power_w · battery_temp_c · soh_pct
- ac_power_w (inverter AC terminals) · meter_power_w (smart meter) · load_power_w · pv_total_power_w
- grid_frequency · power_factor
- Independent CW12x logger CSV: V1/V2/V3 · I1/I2/I3 · P · Q · PF · F · Wh(+) · Wh(-), at 1-second intervals

Appendix C: Sign Conventions

- battery_power_w > 0 : battery is charging (energy flowing into battery)
- battery_power_w < 0 : battery is discharging (energy flowing out of battery)
- active_power_w > 0 : inverter is exporting AC (delivering power)
- active_power_w < 0 : inverter is importing AC (absorbing power)
- grid_power_w (meter, CW12x convention) > 0 : importing from grid; < 0 : exporting to grid

Appendix D: Contact

For questions about the test methodology, data, or any aspect of this report:

Name	Glen Morris
Company	Smart Energy Lab
Email	glen@smartenergylab.com.au
Test instrumentation source	https://github.com/glenmo/foxess_efftest